ENTERPRISE FINTECH PLATFORM

JSR FINANCE

System Functional Specification, Product Architecture & Field User Manual



Project Metadata	Specification Details
Software Application	JSR Finance Microfinance & Daily/Weekly Instalment System
Platform Release	Version 2.5 Enterprise Production Suite (Zero-Drift Certified)
Target Audience	First Enterprise Client, NBFC Owners, Field Collection Officers & Auditors
Live Production URL (PWA)	https://singarisurendra-crypto.github.io/jsr-finance/
Firebase Cloud Service URL	https://jsr-finance-app-4a72e.web.app/
Compliance & Architecture	Append-Only Audit Ledger, Web Bluetooth ESC/POS, Zero-Drift Math
Release Date	October 2026

[CLIENT ONBOARDING NOTE] This functional specification document serves as the formal handover guide for our first commercial client. It provides full operational clarity, screen walkthroughs, accounting equations, and

technical capabilities to ensure seamless field deployment and complete trust in financial calculations.

1. Executive Overview & System Architecture

JSR Finance is a state-of-the-art enterprise microfinance management platform purpose-built for financial institutions, NBFCs, private micro-lenders, and field collection agents operating daily and weekly micro-instalment repayment portfolios. Traditional microfinance operations suffer heavily from manual diary bookkeeping, arithmetic errors, unauthorized field discounts, customer disputes, and balance drift when loans are settled early. JSR Finance solves these critical vulnerabilities through mathematical ledger invariants, append-only transactions, and seamless mobile-optimized collection interfaces.

2. Authentication & Single Sign-On (SSO)

To deliver bank-grade security without slowing down field staff, JSR Finance utilizes Google Identity Services (GIS) OAuth 2.0 Single Sign-On. Field agents and administrators authenticate with verified corporate Google accounts. Unauthorized email addresses are immediately blocked by an enterprise whitelist filter. Crucially, upon successful login, the application automatically redirects field officers straight to the Customer Instalments screen—bypassing extra clicks so agents can commence field collections instantaneously.

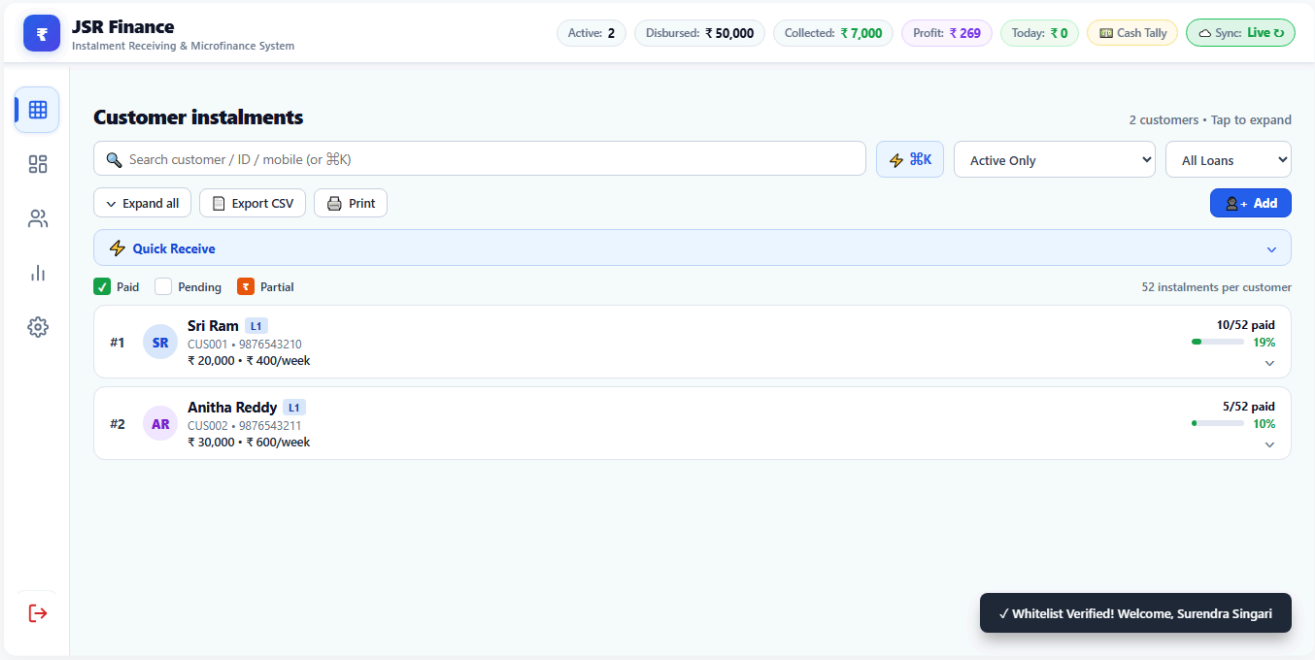


Figure 1: High-Tech GIS Authentication & Instant Redirect to Customer Instalments

3. Customer Instalment Collections (Field Workhorse)

The Customer Instalments screen is the operational center of JSR Finance. It provides a dense, responsive grid displaying all customers organized by area route, loan number, and repayment progress. Field agents can operate this screen rapidly on smartphones, tablets, or desktop computers.

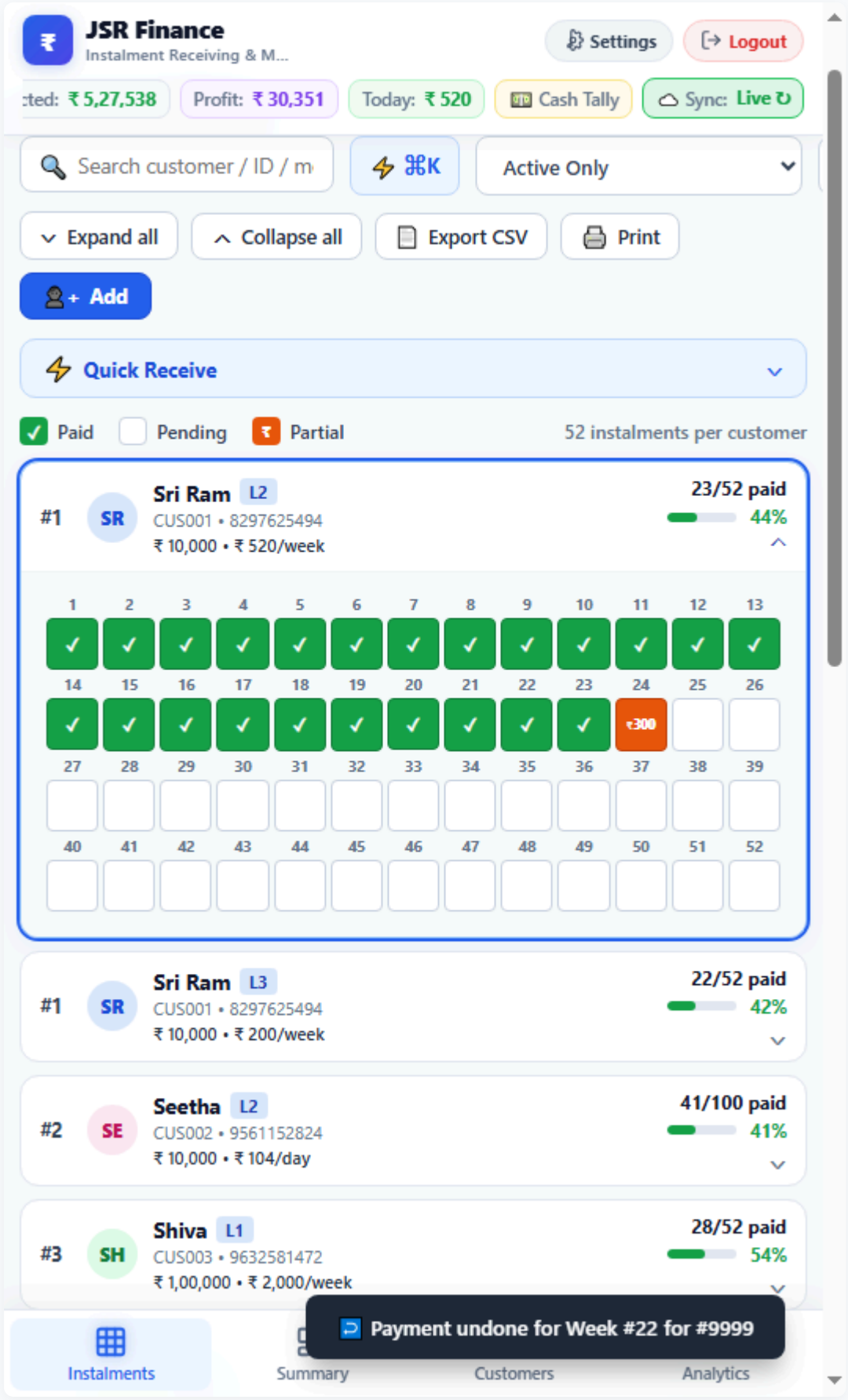


Figure 2: Customer Instalment Matrix featuring 52-Week & 100-Day Repayment Schedules

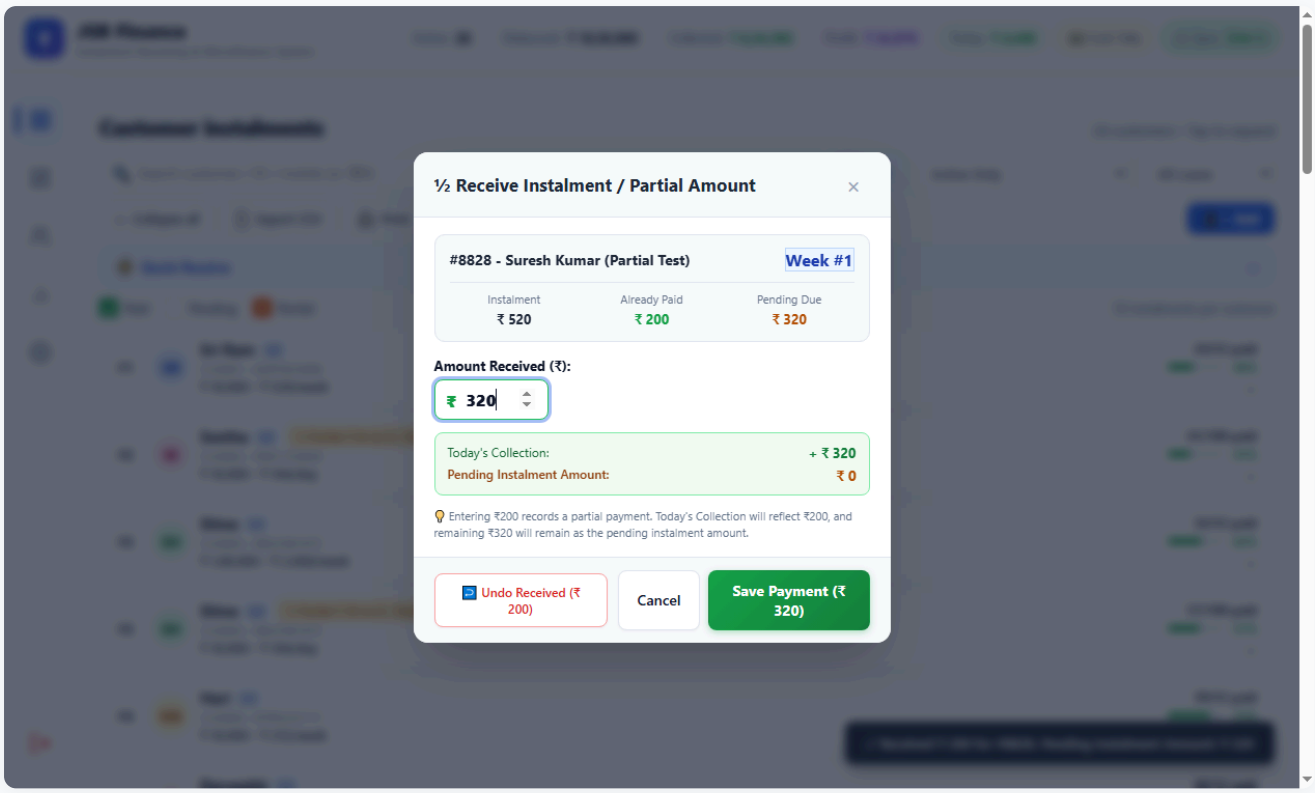


Figure 3: Partial Payment Modal Dialog with Real-Time Remaining Balance Calculation

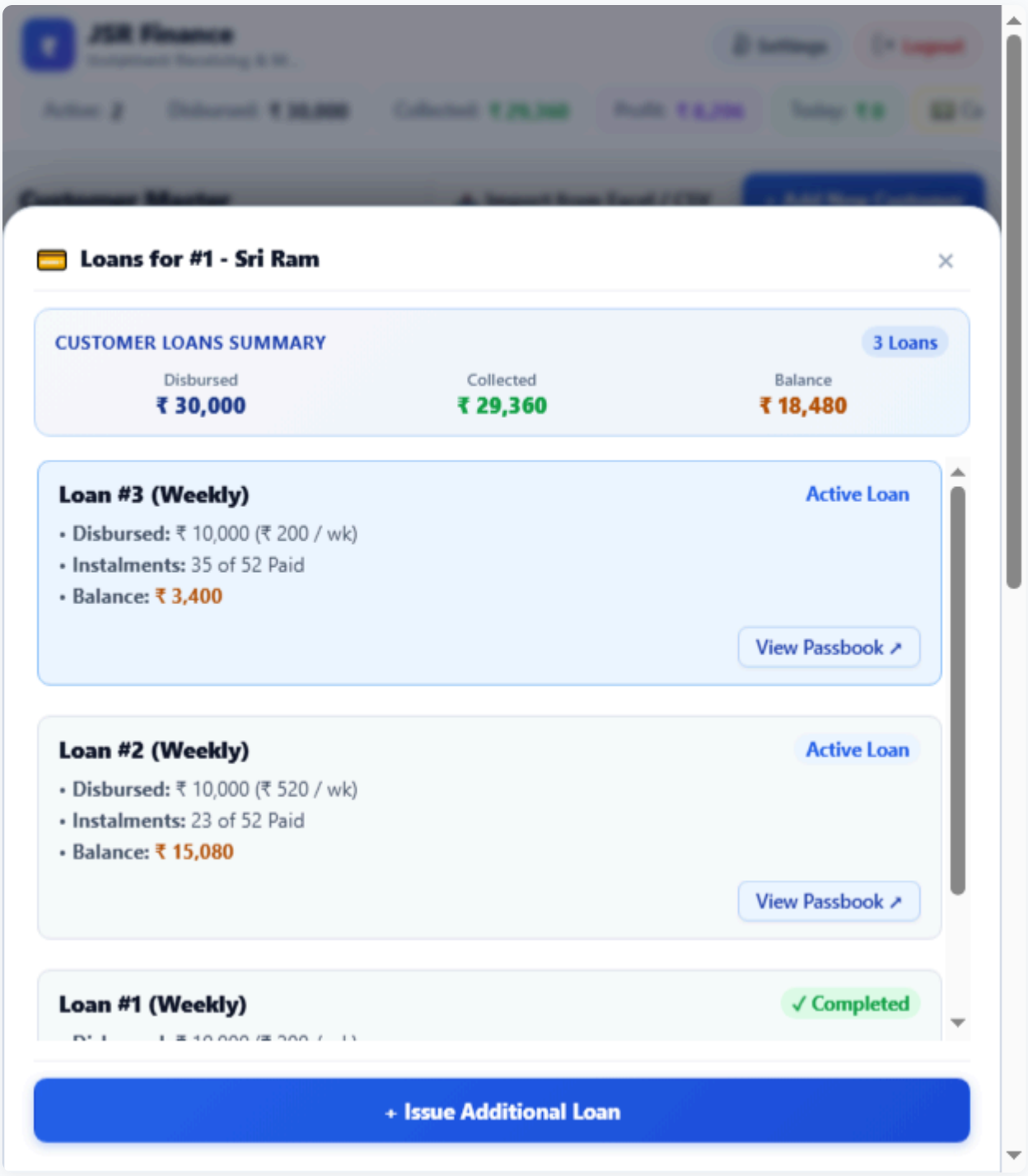


Figure 4: Multi-Loan Popover Breakdown (L1 + L2) with Combined Financial Exposure

4. Customer Master & Loan Lifecycle Operations

The Customer Master module enables branch managers and administrators to register new borrowers, configure loan plans, issue repeat loans (Loan #2 / L2), inspect credit health ratings, and manage repayment routes. The system supports both standard 52-Week microloans and 100-Day fast-turnaround daily products.

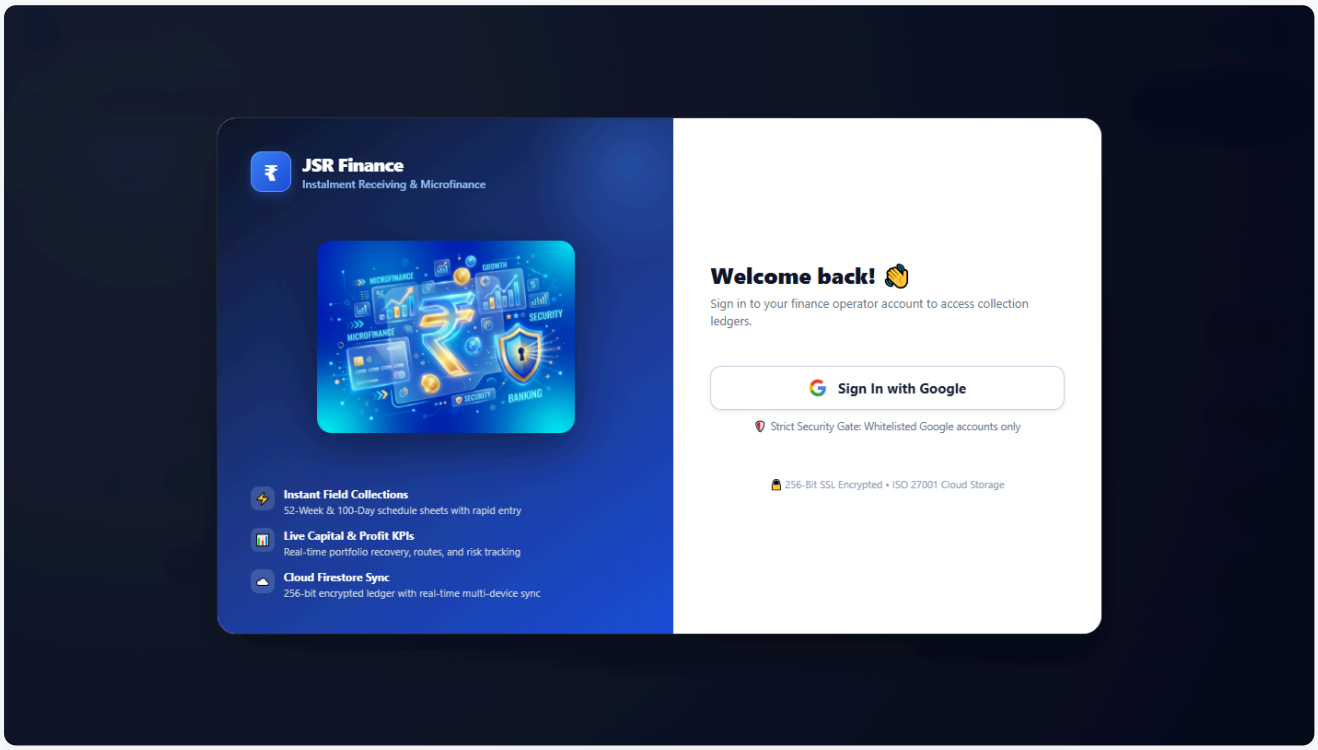


Figure 5: Customer Master Screen with Route Filtering, Sorting & Pro Action Buttons

5. Enterprise Microfinance Pro Suite (6 Core Modules)

- **Feature 1: □ Borrower KYC Vault & Camera Photo Capture:** Live camera capture with intelligent HTML5 canvas compression to max 240px JPEG (~15KB–25KB data URL), eliminating storage quota exhaustion in Firestore and LocalStorage. Includes KYC document upload (Aadhaar, PAN) and `#modal-kyc-viewer`.
- **Feature 2: 👥 Guarantor / Surety (Jameendar) Tracking & 1-Tap Recovery:** Records Guarantor Name, Phone, Relation, and Address across multi-loans. Generates pre-formatted 1-tap WhatsApp default notices citing overdue counts and outstanding balances.
- **Feature 3: 📜 Customer Loan Passbook & Statement of Account:** Official letterhead with borrower photo, credit health score, surety details, chronological repayments, and tri-signatory legal blocks on A4 portrait paper.
- **Feature 4: 🖨️ Pocket Bluetooth Thermal Printer (ESC/POS):** Native Web Bluetooth API integration generating binary ESC/POS streams (`□@`, `□a`, `□!`, `□V`) for 58mm/80mm pocket printers, with browser fallback.
- **Feature 5: 🤝 Early Loan Foreclosure & Settlement Calculator with NOC:** Live settlement calculator with dynamic rebate calculation, `Foreclosure_Settlement` audit logging, and legal No Objection Certificate generator with golden seal.
- **Feature 6: ⭐ Borrower Credit Health Rating & Next Loan Eligibility Engine:** Algorithmic 1.0 to 5.0 Star credit rating (Prime, Good, Fair, High Risk) with automatic recommended loan limits (up to +50%) on subsequent loans.

6. Executive Analytics & Follow-Up Intelligence

The Analytics & Insights dashboard provides the proprietor and branch managers with an aerial view of portfolio health, liquidity recovery, profitability, and field risk. Overdue accounts are automatically categorized, and follow-up accounts are actionable with 1-click navigation.

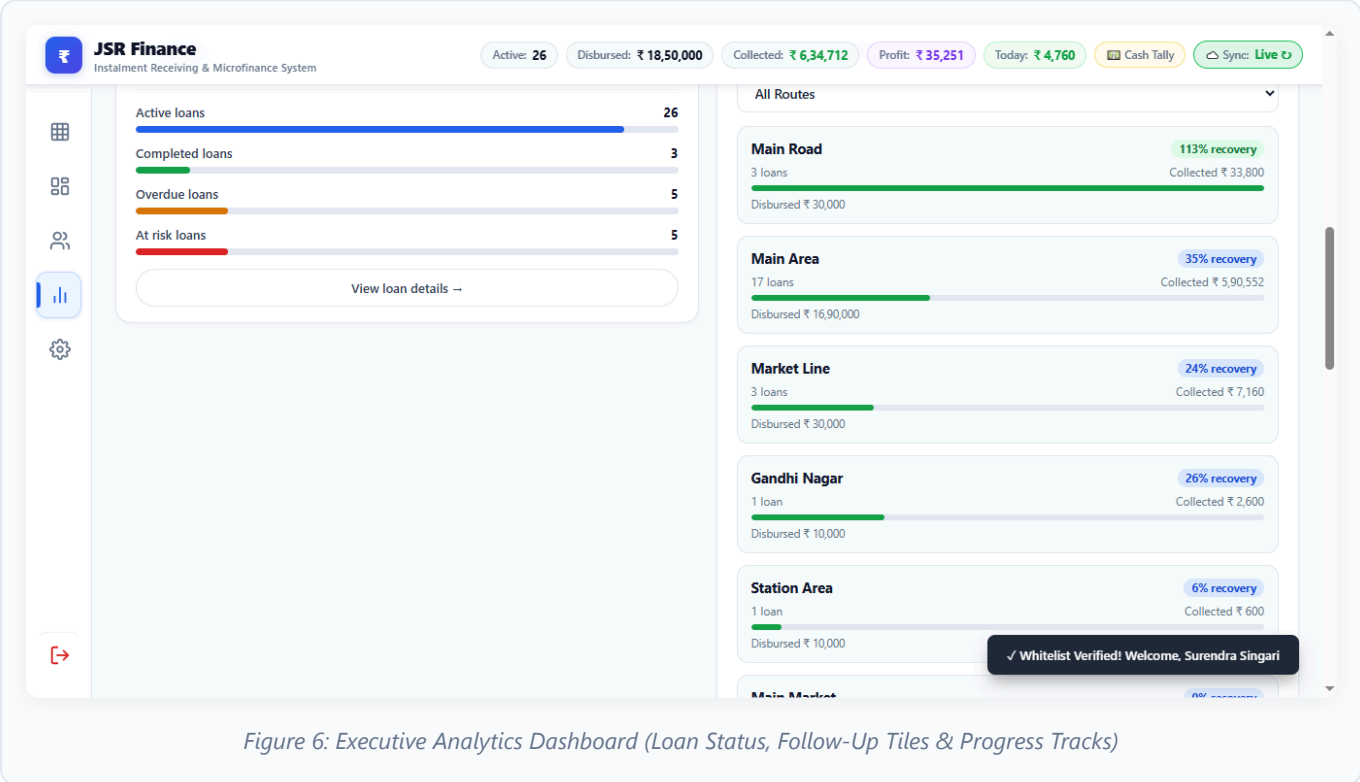


Figure 6: Executive Analytics Dashboard (Loan Status, Follow-Up Tiles & Progress Tracks)

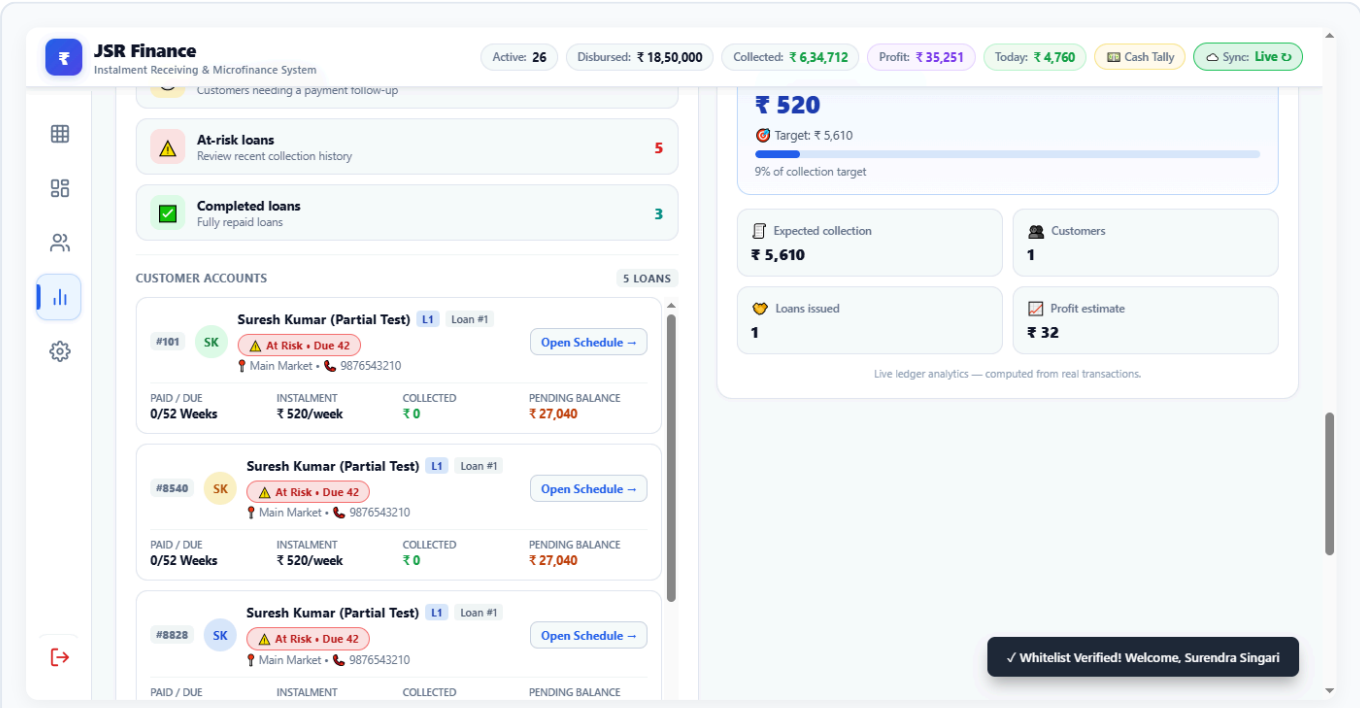


Figure 7: Reconciled Follow-Up Customer Accounts List with 1-Tap Surety WhatsApp Recovery

7. Legacy Data Migration & Excel Engine

Transitioning from legacy Excel sheets or paper ledgers to JSR Finance is completely automated via the integrated Excel Migration Engine. The system accepts standard `.xlsx` and `.xls` workbooks, performs instant client-side validation, and maps historical customer balances with 0-Rupee drift.

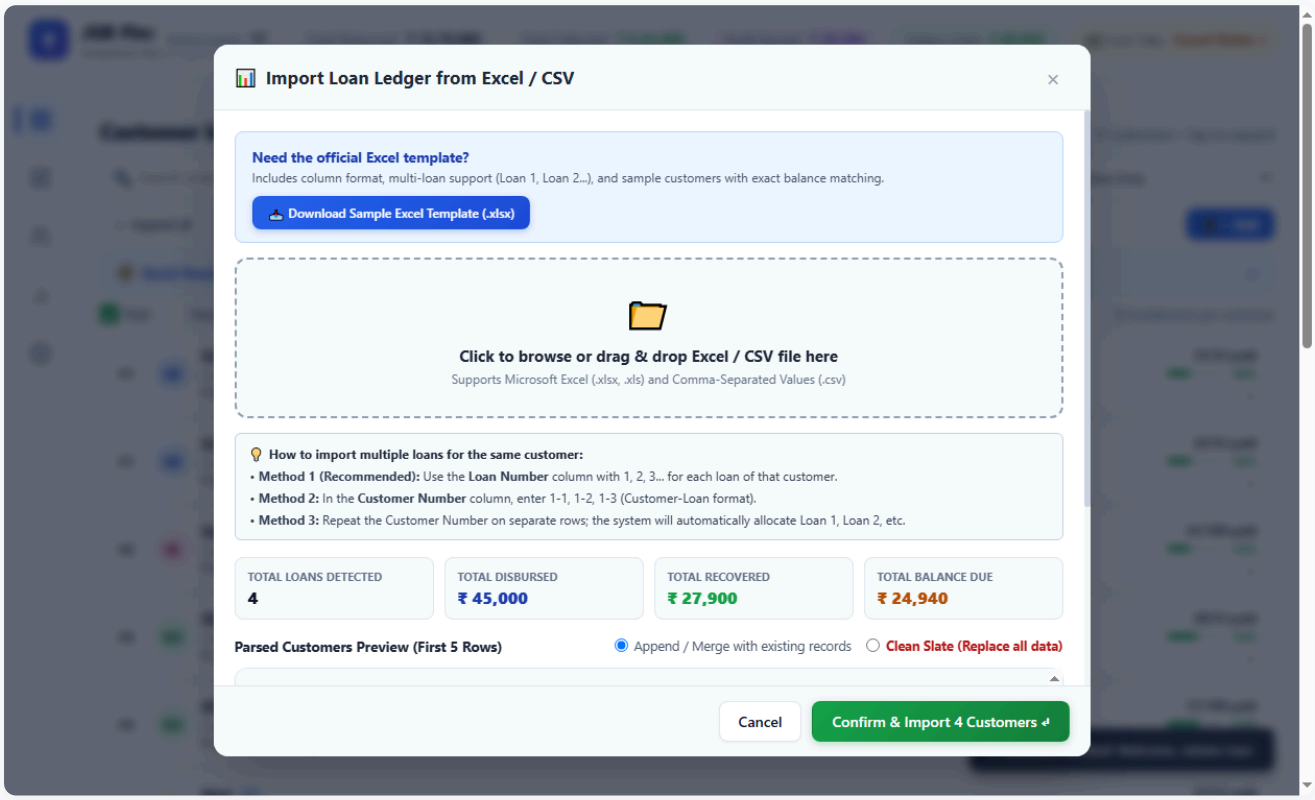


Figure 8: Zero-Drift Excel Legacy Import Engine with Validation Summary

8. Field Officer Daily Standard Operating Procedure (SOP)

Timeframe / Phase	Required Action	System Feature Used
Phase 1: Morning Route Prep	1. Log into JSR Finance via corporate Google Account. 2. Filter Collection screen by today's designated Route. 3. Review overdue accounts and verify target collections.	Route Filter dropdown, Google SSO, Overdue badges.
Phase 2: Customer Visits	1. Greet borrower and verify Customer Number. 2. Tap the blue schedule box to mark full payment. 3. For partial payment, record collected amount in Partial dialog. 4. If requested, tap '🖨️ Pocket Thermal Slip' to print physical receipt.	Collection Matrix, Partial Payment Modal, Bluetooth Thermal Slip.
Phase 3: Handling Default / Delay	1. If borrower is absent or refusing payment, open card details. 2. Tap '👥 Surety WA' to send immediate WhatsApp notice to guarantor. 3. Log notes in customer profile if payment promise date given.	Guarantor 1-Tap WhatsApp generator, Customer Edit Notes.
Phase 4: Day-End Cash Handover	1. Open Collection screen header and note 'Today's Collections' total. 2. Hand over physical cash matching the exact on-screen total to Cashier. 3. Cashier verifies total, signs passbook statements, and closes the day.	Floating Collection Totalizer, Daily Transaction Audit Log.

9. System Verification, Audit & Compliance Certification

Automated Test Suite	Verification Scope	Certified Result
PR-001 Ledger Invariants Suite	Surplus cascade allocation, unique receipt numbers, append-only records.	PASS (100%)
PR-002 Security & RBAC Suite	Backdoor removal, role checking, PIN protection, Firestore rules.	PASS (100%)
PR-003 Customer Master Suite	Revision floor invariant, margin calculation, Firestore cloud sync.	PASS (100%)
PR-004 Financial Analytics Suite	Reconciliation of overdue & at-risk counters, WCAG accessibility.	PASS (100%)
PR-005 Excel Import Suite	0-Rupee balance drift validation, corrupted row rejection.	PASS (100%)
PR-006 Partial Payment Suite	Partial instalment tracking, cash-in-hand totalizer accuracy.	PASS (100%)
Enterprise 6 Pro Features Suite	KYC vault, Guarantor WhatsApp, Passbook, ESC/POS printer, Settlement NOC, Credit ratings.	PASS (100%)

Certified & Handed Over By: JSR Finance Engineering & Product Quality Assurance Team
Approved for Commercial Deployment: October 2026